Isocolon, parison, and comparable-set construal: A corpus test of formal balance

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Abstract

Isocolon is usually discussed as a figure of balanced cola, but its rhetorical interest comes from a stronger idea: balanced form can help two units look comparable. A corpus can't test that reader-response claim directly. It can test a narrower textual claim: if writers use balanced form to present adjacent units as comparable co-members, then measurable formal balance should cluster in the discourse relations that invite that presentation.

This paper tests that claim in the Georgetown University Multilayer corpus (GUM) with enhanced Rhetorical Structure Theory (eRST) annotations. The analysis scores 17,456 adjacent relation pairs from 299 documents across 24 genres. The composite formal-balance score combines three subscores: ISOCOLON, length balance; PARISON, surface syntactic parallelism; and LEXICAL ECHO, lemma-sequence similarity plus content-lemma overlap.

The broad target family of comparison and co-ordination relations scores 4.1 points higher than the non-target baseline on a 0–100 reporting scale, about 0.28 standard deviations. The largest effects come from *joint-list* and *joint-disjunction*, where co-ordination makes parallel form grammatically cheap. The cleaner non-enumerative evidence comes from *adversative-contrast*, which remains positive inside an adversative-only comparison. The main controlled signal is parison.

Length balance appears under mean-only conditioning, but it contributes little to the main composite estimate once maximum length is also controlled. The eRST antithesis label sits near zero, and concession is negative. The result supports a modest textual claim: formal balance, especially parison, belongs to the profile of some comparable-set relations, but relation names don't guarantee classical figures.

1 Introduction

Rhetorical theory has long treated figures as patterned forms with argumentative force. In that tradition, ISOCOLON joins two kinds of evidence that are easy to confuse: formal balance between adjacent

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cola, short rhetorical units in sequence, and a reader's sense that the units invite comparison. The motivating claim is simple. When writers place two clauses or phrases in balanced form, they may present the comparison between them as more explicit, pointed, or forceful.

This paper asks how much of that claim can be tested without a reader experiment. The answer depends on the explanatory level. At the reader-response level, the claim says that isocolon changes what readers notice or how forcefully they construe a relation. Corpus evidence alone can't show that. At the textual level, the claim turns distributional: writers may preferentially use balanced form where they construe two units as comparable. That claim is testable.

The corpus test uses a narrow hypothesis. When a discourse relation presents two adjacent units as comparable co-members, the pair should show more formal balance than pairs expressing cause, condition, elaboration, sequence, or other non-comparative relations. A discourse relation is the annotated link between two stretches of text: for example, one unit may contrast with another, explain it, elaborate it, or continue a sequence. The study uses the Georgetown University Multilayer (GUM) corpus and enhanced Rhetorical Structure Theory (eRST) annotations because they provide genre-diverse texts, discourse-relation labels, and signal annotation (Zeldes, 2017; Zeldes et al., 2025).

I use a constructed composite formal-balance score to make that comparison. For every adjacent pair, the composite formal-balance score measures how much the two units resemble each other in length, surface syntax, and lexical material. It's graded, not binary: high values of the composite formal-balance score indicate strong formal resemblance, not confirmed rhetorical figures.

The analysis uses the eRST relation export: mostly GUM, plus the smaller GENTLE challenge set. Access to the Rhetorical Structure Theory (RST) Signalling Corpus is still pending, so that corpus isn't part of this analysis. Once available, it can support later triangulation against a broader inventory of coherence-relation signals.

The pattern refines the rhetorical claim. Relations in the broad target family are more formally balanced than matched non-target relations, but the largest effects for that family occur exactly where a grammar-driven co-ordination account is strongest: lists and disjunctions. Ordinary adversative contrast is smaller but cleaner, because it survives inside an adversative-only comparison. Antithesis alone is near zero, and concession is negative. The pattern suggests that formal balance, especially parison-like syntactic parallelism, tracks some acts of comparative construal better than it tracks the inherited relation name ANTITHESIS.

2 Rhetorical figures as form-function pairings

The analysis starts from Fahnestock's account of rhetorical figures as form-function pairings. In her work on scientific argument and rhetorical style, figures are recurrent formal arrangements that can epitomize lines of reasoning (Fahnestock, 1999, 2011). That view gives the corpus prediction. If a figure serves an argumentative operation, then the figure should be distributed unevenly across the discourse contexts where that operation is being performed.

Fahnestock's discussion of the figures of argument speaks directly to this project. ISOCOLON concerns approximate equality in the length of adjacent cola, while PARISON concerns similarity of grammatical structure.

The two often co-occur, but the properties differ. A pair can be close in length without being syntactically parallel, and it can be syntactically parallel without being closely balanced in length. The classical tradition links both to comparison and contrast because similar form can make two units cohere as an argumentative set (Fahnestock, 2011).[1]

A figure can help make a relation salient without being obligatory whenever the relation is present. For the corpus test, uneven distribution is expected, and frequency by itself gives no direct measure of the rhetorical value of the figure.

The same point motivates computational work on rhetorical figures. Harris and Di Marco (2017) argue that figural form is often iconically related to discourse function, so surface patterning can provide evidence about pragmatic and argumentative actions. Harris et al. (2018) develop an annotation scheme for figures in part because figure identification requires attention to both formal regularity and rhetorical role. Lawrence et al. (2017) make the parallel point for argument mining: rhetorical figures can identify regions where argumentative work is being done, but only if formal detection is connected to the content and relation being advanced.

Together, these studies motivate the working assumption. The analysis pairs surface balance with a discourse relation that makes two units available for comparison. The operational question asks whether formal balance, separated into isocolonic length balance, parison-like syntactic parallelism, and lexical echo, is higher in the relations where rhetorical theory predicts that balance has value.

3 From candidate form to rhetorical function

The primary methodological challenge concerns over-attribution. Repetition and parallelism are common in ordinary text, and only some instances have the rhetorical force that figure theory cares about. Work on automatic chiasmus detection makes the problem vivid. Dubremetz and Nivre (2018) show that trivial repetition patterns are easy to find, but repetitions with rhetorical effect are much harder to separate from accidental or irrelevant patterns. They treat figure detection as a ranking problem instead of a clean binary classification problem. The problem is not finding parallelism. The problem is persuading it to mean something.

That lesson carries over to isocolon and parison. Exact repetition, list formatting, dialogue formulas, quoted slogans, and segmentation artifacts can all look formally balanced. Some are rhetorically relevant, but many aren't instances of the target phenomenon. Recent survey work makes the broader point: rhetorical figure detection remains limited by small datasets, inconsistent definitions, and over-reliance on rule-based surface cues (Kühn & Mitrović, 2024; Kühn et al., 2024). A credible corpus test should use surface measures cautiously, validate them against human judgment, and avoid treating high balance as equivalent to a gold figure label.

The analysis below follows that strategy. The composite formal-balance score is a continuous outcome in the distributional analysis and a ranking score in validation. It ranks adjacent relation pairs by how much measurable formal balance they display. The qualitative audit then asks whether top-ranked examples are plausible formal-balance candidates after excluding exact repetition, dialogue formulas, and other false-positive types. This narrower aim is the right target for a first rhetorical-function test; automatic figure recognition would be a different task.

[1] I approach these terms as a corpus linguist with recent rhetorical reading. That outsider position matters: the Greek terms motivate hypotheses, but the corpus tests the mappings. Less formally, this study stays suspicious of its own Greek.

4 Corpus and measurement

4.1 The corpus and what counts as a pair

The corpus comes from the English Georgetown University Multilayer corpus with enhanced Rhetorical Structure Theory annotations (GUM/eRST). GUM provides genre-diverse, richly annotated English texts (Zeldes, 2017). eRST extends RST-style discourse annotation with graph structures and explicit signal annotation across spoken and written genres (Zeldes et al., 2025). The DISRPT discourse-parsing shared-task export used here is a standardized release prepared for computational discourse parsing. It also includes a smaller GENTLE challenge set. GENTLE is an out-of-domain test collection distributed with the same export, not a second main corpus.

Sentence adjacency alone would be too coarse. Two neighbouring units can contrast, elaborate, narrate, repair, or list, and the hypothesis depends on which relation holds between them. The project also needs enough genre variation to separate a rhetorical effect from a register-specific habit.

The analysis uses adjacent discourse-relation pairs. The dataset contains 17,456 such pairs from 299 documents across 24 genres. These discourse units serve as candidate cola-like spans, with no claim that they're classical cola in a strict prosodic sense.

Adjacency matters because the rhetorical claim is local. The analysis asks whether two neighbouring units in a relation are formally balanced. It doesn't ask whether two distant units in the same document resemble each other.

A relation enters the analysis if eRST marks it as explicit or implicit, both spans have token intervals, and the two intervals are adjacent in document order. A token interval is the start and end position of a unit in the document's token sequence. An explicit relation has a marked signal in the text, such as a connective or other cue; an implicit relation is inferred by the annotator without such a signal. Non-adjacent linked spans are excluded. Concurrent graph relations are kept as pairwise relation rows when their spans meet this adjacency criterion.

The extracted set contains 16,278 rows with collection label GUM and 1,178 with collection label *GENTLE*; the 24 genres are the document-genre labels in the extracted files. The derived DISRPT relation table has relation rows and token intervals, not the full eRST discourse graph, so the extraction diagnostics are interval-level checks rather than a complete graph-theoretic audit. Appendix A reports the full extraction counts.

4.2 Three measures of formal balance

For each pair, I computed three surface measures. First, ISOCOLON measures length balance over token, character, and estimated syllable length. Second, PARISON measures similarity in dependency and lexical-category profiles. Third, LEXICAL ECHO measures repeated or closely corresponding lexical material. The composite formal-balance score combines the three subscores as a graded measure of formal balance. The composite formal-balance score gives evidence about formal balance, not a figure label for the pair.

A token is the corpus's word-like unit. For instance, *Get your feet off of there* has six word tokens under ordinary tokenization. Token counts are useful for isocolon, but they aren't the whole measure because short units can match in length while differing sharply in structure.

The subscore calculations stay deliberately simple. For any two positive values n_1 and n_2, the component-balance value $B(n_1, n_2)$ is

$$B(n_1, n_2) = 1 - \frac{|n_1 - n_2|}{\max(n_1, n_2)}. \tag{1}$$

The value is 1 when the two units match exactly on that dimension and falls toward 0 as they diverge. Two units of 8 and 10 words, for instance, score $1 - |8 - 10|/10 = 0.8$ under Equation 1. Two units of equal length score 1. The isocolon subscore averages this quantity over word tokens, characters, and estimated syllables.

The parison subscore compares the two units' grammatical make-up rather than their wording. It averages edit similarity over three sequences, each read in token order: the coarse Universal Dependencies lexical-category field (*UPOS*), the language-specific lexical-category field (*XPOS*), and the dependency relations, which name the grammatical function (subject, object, modifier, and so on) that ties each word to the word it depends on.

Edit similarity counts how few small changes, inserting, deleting, or swapping one item, turn one such sequence into the other, rescaled to lie between 0 and 1 (Levenshtein, 1966). Two units that run through the same categories in the same order score near 1. Because the sequences are read flat, left to right, the measure registers shared category order but not the branching shape of the sentence; it's a surface profile, not a comparison of full parse trees.

For a schematic example, compare *The dog barked* with *A cat slept*. Their wording differs, but their lexical-category sequence is the same: article, noun, verb. Their dependency-relation sequence also follows the same surface pattern. The parison subscore is high because the grammatical profile matches, even though the lexical-echo subscore is low. By contrast, a pair such as *The dog barked* and *After dinner* has little category or dependency alignment and gets a low parison value.

The lexical-echo subscore applies the same comparison to the words themselves, in their lemma form, the dictionary form that counts *run*, *runs*, and *running* as one word. It looks only at content words, the meaning-bearing nouns, verbs, adjectives, and adverbs, setting function words such as *the* or *of* aside. Two quantities are averaged: edit similarity over the content lemmas in order, and Jaccard overlap, the share of content lemmas the two units have in common regardless of order (Jaccard, 1901).

The GUM/eRST files use the Conference on Natural Language Learning format for Universal Dependencies (CoNLL-U), which records tokenization, lemmas, lexical-category tags, and dependency labels. Punctuation tokens are excluded from the word, syntax, and lexical sequences, while colon and semicolon position are kept as separate controls. Syllable counts are approximate vowel-group counts. For discontinuous units, length is counted from the tokens assigned to the unit, not from the full interval width including intervening material.

The composite formal-balance score S_{formal} is defined in Equation 2.

$$S_{\text{formal}} = 0.40 \times \text{isocolon} + 0.45 \times \text{parison} + 0.15 \times \text{lexical echo}. \tag{2}$$

These weights define only the composite formal-balance score. The subscores are also reported separately, so the analysis can distinguish isocolonic length balance from parison-like syntactic parallelism and lexical echo.

The weights make syntax and length the main ingredients and keep lexical echo smaller, because repeated words can mark balance but can also mark plain repetition. A later robustness check repeats the analysis across many alternative weightings, so the argument doesn't depend on this exact 0.40/0.45/0.15 split.

4.3 Defining the target relations

The composite formal-balance score is only a property of each adjacent pair. It says how closely the two units match in length, surface syntax, and lexical material. It doesn't say that the pair belongs to a target relation, and it doesn't by itself identify a rhetorical figure. The target definitions below do separate work: they say which eRST relation labels should, by hypothesis, tend to have higher values of the composite formal-balance score.

The analysis then sorts the eRST relation labels into hypothesis groups. Table 1 gives the mapping. These groups are analysis groups, not new annotations: they say which existing eRST labels count as targets for this test. The broad target family includes *adversative-antithesis*, *adversative-contrast*, *joint-disjunction*, and *joint-list*. This family tests whether formal balance rises in relations that place units in a comparable or co-ordinated frame. The narrow adversative family includes only *adversative-antithesis* and *adversative-contrast*.

Individual-relation analyses separate the effect for the broad target family into its individual labels.

The label prefixes matter. *adversative-** labels mark contrastive or concessive relations. *joint-** labels mark multinuclear relations, where the linked units have equal discourse status. The label *adversative-antithesis* is an eRST relation name; it isn't already a hand annotation of the classical figure ANTITHESIS.

Analysis role	Labels, rationale, and caveat
Broad target family	*adversative-antithesis*; *adversative-contrast*; *joint-disjunction*; *joint-list*. Contrast, alternatives, and co-enumeration place units into a shared frame. The joint labels are co-ordination-rich and also the strongest grammar-driven alternative explanation.
Narrow adversative family	*adversative-antithesis*; *adversative-contrast*. Tests whether the broad result survives without list and disjunction labels.
Cleaner non-enumerative test	*adversative-contrast*. Tests contrast against other adversative relations, where list syntax is not the main explanation.
Contrastive control	*adversative-concession*. Adversative but often asymmetric; predicted to be weaker or negative.
Surface positive control	*restatement-repetition*. Expected to score high by form alone; used as a false-positive diagnostic, separate from the target hypothesis.

Table 1: Mapping from eRST relation labels to analysis roles. The table defines the operational grouping. Independent annotation of comparable-set construal remains future work.

4.4 Estimating the effects

The primary comparisons ask a plain difference question: after removing background differences that could distort the comparison, do target relations receive higher values of the composite formal-balance score than other relations? The controls remove differences that can make the comparison too easy: document, explicit signalling, sentence adjacency, punctuation, mean length, and maximum length. Document controls mean that, where possible, target and non-target pairs are compared inside the same text.

For readability, I report effects as score-point differences on a 0–100 version of the composite formal-balance scale, where a pair sits near 0 when its two units share little measurable form, with different lengths, structures, and words, and near 100 when they match closely on length, syntax, and wording. The underlying scores range from 0 to 1. Differences between two groups can range from -1 to $+1$ on the raw scale, or from -100 to $+100$ after the score-point rescaling.

Figure 1 shows the empirical range behind that scale and splits it by target status. The 0–100 scale is a readability transformation, not a claim that the observed data fill the whole interval evenly. In the 17,456 adjacent pairs, the median composite value is 32.8. The middle half runs from 22.3 to 41.9, and 95% of pairs fall below 55.9. A few exact or near-exact matches reach the upper tail, but most of the empirical scale is the lower and middle part of the formal-balance range. The broad target family is visibly shifted right, but the two distributions overlap heavily.

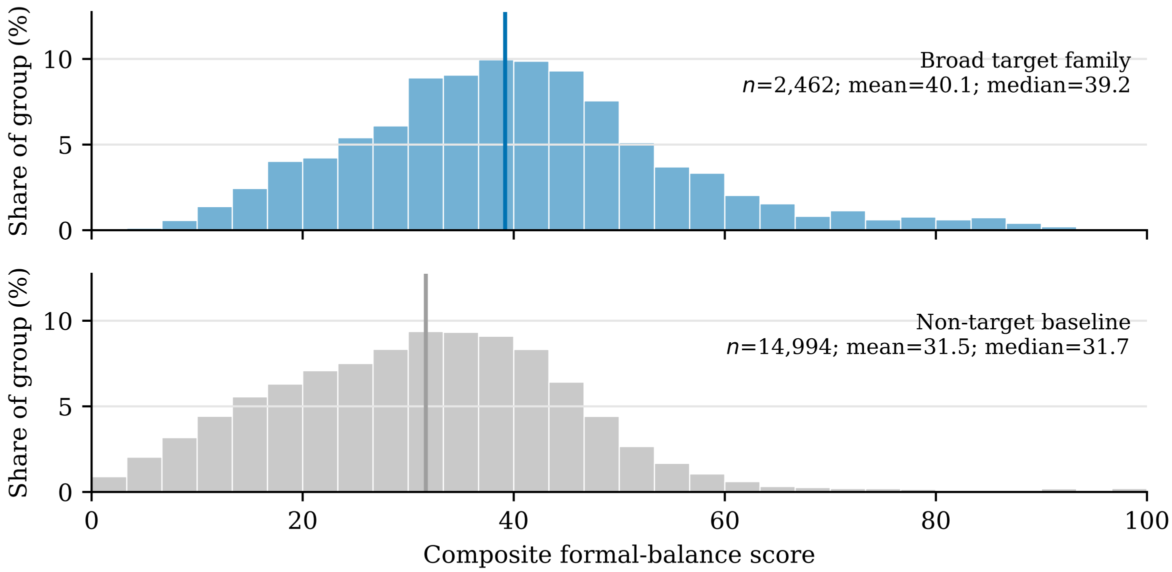

Figure 1: Distribution of the composite formal-balance score after rescaling to 0–100, split by target status: broad target family versus non-target baseline. Bars show percentages within each group, not raw counts, so the larger non-target baseline doesn't dominate the display. Vertical lines mark group medians. The plot is descriptive; the 4.1-point effect reported below is adjusted for document, length, punctuation, and other controls.

For scale, the most extreme comparison isn't a target relation at all. It is simple restatement or repetition: after controls, repeated pairs sit 19.0 points above other pairs. One such pair, *Get your feet off of there.* followed by *Get your feet off of there.*, receives a composite formal-balance score of 100.0. That's 68.3 points above the non-target median of 31.7. Table 2 gives additional anchors for the raw per-pair score scale, with one target example and one non-target example near each anchor.

Score	Status	Gap	Unit 1		Unit 2
100.0	Non-target	+68.3	*Get your feet off of there.*	||	*Get your feet off of there.*
95.2	Target	+63.5	*You can use this on twist outs.*	||	*You can use this on braid outs.*
60.0	Non-target	+28.3	*I got to go out on the road.*	||	*I want to get my chops back up.*
59.9	Target	+28.2	*I worked with some great musicians*	||	*and performed around the world.*
50.0	Non-target	+18.3	*it forces you to take action,*	||	*as you can't afford to waste any time.*
50.0	Target	+18.3	*the house festooned with flowers*	||	*and stocked with coconut liquor.*
40.0	Non-target	+8.3	*Steven measured and mixed the plaster of Paris*	||	*and prepared for the cast*
40.0	Target	+8.3	*maybe I should vlog today*	||	*and just see how my day turns around*
30.0	Non-target	−1.7	*For your joke to be funny,*	||	*it has to surprise the audience in some way,*
30.0	Target	−1.7	*Shorts and sleeveless shirts are highly inappropriate,*	||	*as are short skirts.*
20.0	Non-target	−11.7	*Yeah no.*	||	*Not a good look.*
19.6	Target	−12.1	*We have no choice*	||	*but to return to direct negotiations between the two parties.*

Table 2: Example raw composite formal-balance scores. Gap is measured in score points against the non-target median of 31.7. TARGET means the pair belongs to the broad target family; NON-TARGET means it doesn't. The || symbol marks the boundary between the two units. These examples calibrate the raw scale; they aren't adjusted effects.

The gap between the broad target family and the non-target baseline is 4.1 points, about a fifth of that repetition gap. The empirical standard deviation of the composite formal-balance score is 14.8 points, so the gap for the broad target family is about 0.28 standard deviations. The effect is a small distributional shift. It doesn't separate target from non-target pairs cleanly, as Figure 1 already shows. It remains worth modelling because the estimate stays positive after accounting for related pairs within documents and under the robustness checks below.

I estimate these differences with ordinary least squares regression. In plain terms, it reports the average gap in balance between target and non-target pairs that are otherwise alike, holding the controls (document, length, punctuation, sentence adjacency, and explicitness) constant, so the gap isn't just a by-product of those background differences.

The estimates are descriptive adjusted differences. A positive value means that the target relation group has a higher composite formal-balance score than its comparison group after the controls. A negative value means the target group is lower.

For the composite formal-balance score, longer spans have more chances for structural mismatch, so the main models control for both mean span length and maximum span length. The isocolon subscore needs different treatment because its token component is built from those same quantities. If $m = \max(n_1, n_2)$ and $\mu = (n_1 + n_2)/2$, the token-length component in Equation 1 can also be written as $2\mu/m - 1$. A model that controls for both μ and m holds fixed the ingredients of the outcome and leaves little token-length balance to estimate. The same concern applies to the character and syllable components. For that reason, the isocolon models control for mean length but omit maximum length.

The models control for explicit relation status, same-sentence adjacency, colon punctuation, semicolon punctuation, standardized log mean length, standardized log maximum length, and document fixed effects. Document fixed effects are document-specific baselines; they compare target and non-target pairs inside the same text where the data allow it. The two length terms enter logged and standardized: taking the logarithm keeps a few very long spans from dominating, and standardizing puts mean and maximum length on a common scale. Document-clustered intervals treat pairs from the same document as related observations; these are the uncertainty ranges reported in the tables (Cameron & Miller, 2015).

A genre-varying analysis asks whether the effect comes from only one or two genres. It first estimates the same adjusted effect separately within each genre. Those single-genre figures are shaky wherever a genre contributes few pairs, so a second shrinkage step pulls each one toward the shared average (Efron & Morris, 1973). Estimates built on little data move the most; well-supported estimates hardly move. The same step reports how far the genres still spread around that shared average once the correction is made.

Robustness checks add mean-only length models, joint-only and adversative-only comparisons, GUM-only and continuous-span-only subsets, leave-one-genre influence checks, composite-weight sensitivity checks, and stratified permutation checks. In the permutation checks, the target labels are shuffled inside matched strata. The question is whether effects this large appear when relation labels are randomly assigned within comparable document, length, punctuation, and explicitness groups.

5 Results

The score-scale discussion above and Figure 1 showed the main descriptive pattern: target and non-target pairs overlap heavily, but the broad target-family distribution is shifted to the right. The adjusted model estimates that shift. On a 0–100 version of the composite formal-balance score, pairs in the broad target family are 4.1 points higher than pairs outside that family after controls. That equals 0.041 on the raw 0–1 scale. The empirical standard deviation of that score is 14.8 points, so the adjusted gap is about 0.28 standard deviations: a little more than a quarter of the ordinary spread in pair scores.

The document-clustered uncertainty interval runs from 3.6 to 4.7 points. The shift is modest, but its estimated direction is stable: the uncertainty interval stays away from zero.

The subscores show what kind of balance carries the effect. The parison subscore difference is 7.7 points, compared with 2.4 points for the lexical-echo subscore. With mean-length control only, the broad isocolon subscore difference is 6.6 points. Table 3 reports the composite formal-balance score and the three subscores separately.

Two columns in that table estimate the same isocolon subscore. They differ only in the length controls. The full-controls isocolon column includes both mean and maximum span length, matching the composite model. That estimate is deflated, 0.8 points for the broad target family, because the model has mostly controlled away the ingredients of the isocolon subscore. The mean-only isocolon column drops the maximum-length control. Read that column as the substantive length-balance result: 6.6 points for the broad target family.

The result for the broad target family is an omnibus check. Its strongest labels are also the labels where co-ordination makes parallel form easiest, so the cleaner rhetorical test comes from more focused comparisons. The narrow adversative family, *adversative-antithesis* plus *adversative-contrast*, shows a smaller 1.3-point composite difference, with an interval from 0.5 to 2.1 points.

The individual labels explain the difference. *joint-disjunction* has an adjusted composite formal-balance effect of 4.8 points, *joint-list* has an adjusted effect of 4.4 points, and *adversative-contrast* has the smaller but more diagnostic effect of 2.4 points. *adversative-antithesis* is near zero at 0.4 points, with an interval that crosses zero, so the estimate is compatible with no adjusted difference. I read this as a mismatch between a discourse-relation label and the classical figure. It gives no evidence against antithesis as a rhetorical resource. *adversative-concession* is negative at -1.3 points, meaning that these pairs are less formally balanced than their comparison baseline after controls.

The relation-level scan also identifies an important positive control. *restatement-repetition* has the strongest adjusted composite formal-balance effect, 19.0 points, with especially large lexical-echo and parison effects. That's exactly what the composite formal-balance score should find: exact identity is highly balanced, with lexical echo stronger than the parison effect. The human audit reported in the next section shows why it needs separate treatment. Repetition can signal clarification, identity, repair, emphasis, or other rhetorical work; exact repetition remains distinct from balanced comparison. The analysis treats restatement/repetition as a control and diagnostic, separate from the target hypothesis.

Genre-varying estimates point in the same direction. After pooling, the mean effect for the broad target family is 3.9 points across 20 genres. The *joint-list* estimate is 4.0 points across 17 genres, and *adversative-contrast* is 2.9 points across the genres where there's enough evidence to estimate it. The estimate for the narrow adversative family is 1.2 points.

The genre-by-genre effects scatter only narrowly around those averages: their spread, one standard deviation, is about 1.1 points for the broad target family, 1.3 for *joint-list*, 0.9 for the narrow adversative family, and 0.7 for *adversative-contrast*. Because that scatter is small next to the effects themselves, the effect for the broad target family isn't the work of one or two genres but recurs across many.

Target	Target n	Other n	Δ composite	Δ isocolon full controls	Δ isocolon mean-only controls	Δ parison	Δ lexical
Broad target family	2,462	14,994	4.1	0.8	6.6	7.7	2.4
Narrow adversative family	605	16,851	1.3	-0.1	4.7	2.4	1.7
joint-list	1,690	15,766	4.4	1.0	6.5	8.2	2.0
joint-disjunction	167	17,289	4.8	0.9	3.6	8.6	4.0
adversative-contrast	262	17,194	2.4	-0.0	7.3	4.5	2.8
adversative-antithesis	343	17,113	0.4	-0.2	2.5	0.8	0.9
adversative-concession	679	16,777	-1.3	0.1	1.1	-2.7	-0.7
restatement-repetition	321	17,135	19.0	4.6	17.0	27.3	32.5

Table 3: Selected adjusted differences on the composite formal-balance score and its subscores, reported as score points. The Δ symbol marks each numeric column as a target-minus-comparison difference, not a raw score. The underlying scores range from 0 to 1; the table multiplies differences by 100, so possible values run from -100 to +100. Positive values mean the target rows score higher than their comparison rows; negative values mean they score lower. Target n gives the number of target rows; other n gives the number of rows outside that target. Isocolon is length balance, parison is syntactic parallelism, and lexical echo is lemma-sequence similarity plus content-lemma overlap. Composite, full-controls isocolon, parison, and lexical estimates use document fixed effects and controls for explicitness, sentence adjacency, punctuation, mean length, and maximum length. The mean-only isocolon column uses the same length-balance outcome but omits maximum length because mean and maximum length nearly determine the isocolon subscore. For that reason, the composite formal-balance score isn't the weighted sum of the displayed mean-only isocolon, parison, and lexical-echo columns. The restatement row is a positive control, separate from the target hypothesis.

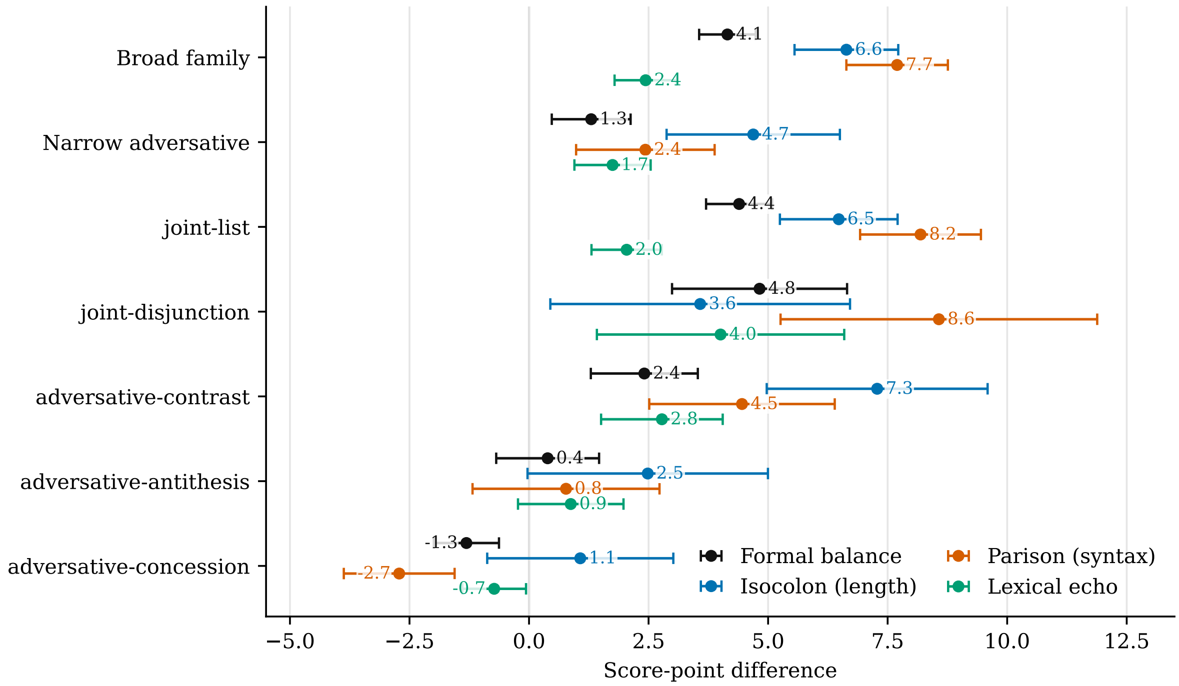

Figure 2: Adjusted score-point differences for the composite formal-balance score, the ISOCOLON subscore, the PARISON subscore, and the LEXICAL ECHO subscore. Points are estimates; horizontal intervals are document-clustered 95% intervals. Effects are shown as score-point differences on a 0–100 scale. The ISOCOLON estimates use mean-length controls and omit maximum length; the other estimates use the main control specification.

5.1 Robustness checks

The main pattern survives the checks that address the obvious alternative explanations. Document-clustered intervals leave the broad target family at 4.1 points [3.6, 4.7]. Mean-only length models recover isocolonic length balance for the broad target family, 6.6 [5.6, 7.7], without conditioning on maximum length. A joint-only comparison shrinks the list/disjunction effect to 1.7 [0.6, 2.8], showing that co-ordination explains a substantial share of the broad result but not all of it.

The adversative-only comparison removes the list/disjunction cases from the baseline and compares *adversative-contrast* with other adversative relations. In that comparison, *adversative-contrast* stays positive at 3.0 [1.4, 4.7] on the composite formal-balance score. GUM-only, continuous-span-only, and leave-one-genre checks give the same qualitative pattern. Stratified permutation checks place the observed effects for the broad family, list/disjunction, and adversative contrast beyond the shuffled-label nulls. A weight-sensitivity check also leaves the broad target-family result positive under all 66 alternative nonnegative weightings. Appendices D and E report the full robustness table, permutation nulls, and weight-sensitivity details.

These checks address the stability of the estimated effects. The next section asks two different questions: whether the composite formal-balance score ranks annotated parallelism cases near the top, and whether top-ranked examples survive rhetorical inspection.

6 Score validation and rhetorical-function audit

The score validation asks whether the composite formal-balance score ranks useful cases near the top. It doesn't ask whether the composite formal-balance score can label figures automatically. The eRST *syn-prl* signal labels mark cases where annotators identified syntactic parallelism as a discourse signal; the label name abbreviates *syntactic parallelism*. Those labels give a small check on formal parallelism, not on rhetorical force.

I report the area under the receiver-operating characteristic curve (AUC) because it has a plain ranking interpretation that needs no statistical background (Hanley & McNeil, 1982). Imagine drawing two pairs at random: one that annotators marked with a *syn-prl* parallelism signal, and one they left unmarked. The AUC is the chance that the composite formal-balance score is higher for the marked pair than for the unmarked one.

A ranking that sorted the two no better than a coin toss would give an AUC of 0.5; a ranking that always put the marked pair on top would give 1.0. So the AUC of 0.73 reported here means that, in this head-to-head, the marked pair wins about 73 times in 100. The measure takes its name from a curve that plots recall against the false-positive rate at every cutoff, but the simple ranking reading is all that's needed here.

On that scale, the composite formal-balance score has an AUC of 0.73, with an approximate Hanley–McNeil interval from 0.66 to 0.80. That interval is the standard large-sample uncertainty range for an AUC (Hanley & McNeil, 1982), not a new effect size. The parison subscore is slightly higher at 0.75 [0.69, 0.82], the lexical-echo subscore is 0.72 [0.65, 0.79], and the isocolon subscore is 0.65 [0.58, 0.72] (Table 5).

The ranking also answers a practical audit question. At the 95th percentile of the composite formal-balance score, the inspected slice has 876 rows; 27 carry *syn-prl*. That's a 3.1% hit rate, about seven times the corpus base rate of 0.4%, and it recovers 37% of all *syn-prl* rows in the corpus. At the 97.5th percentile, tied scores included, 24 of 438 inspected rows carry *syn-prl*. That's a 5.5% hit rate and recovers 33% of all *syn-prl* rows.

Those numbers look poor for automatic labelling but adequate for ranking rare candidates for audit. The validation also stays narrow: the *syn-prl* labels validate a surface parallelism signal. In the eRST source, these are relation-level signals of parallel syntactic construction, with token ranges marking the signal material. Those ranges often cover material in both linked units, but some cover one span or several spans. In this implementation, a pair is positive if either unit contains a *syn-prl* signal in the eRST export, so the validation is a proxy for relation-level syntactic parallelism rather than an exact test of cross-unit alignment. It doesn't validate rhetorical function itself. Appendix B gives the full AUC table and precision/recall figure.

The qualitative audit supplies a pilot rhetorical-function check. I coded 80 candidate pairs in a single-author, non-blind audit; row strata and composite formal-balance values were visible. Automated reviews only flagged borderline cases after coding. The audit isn't interrater-reliability evidence. It identifies false-positive classes and illustrative examples.

The audit doesn't estimate corpus-wide precision. The rows were sampled to stress-test likely successes and likely failures: high target cases, repetition controls, adversative cases, and non-target artifacts. The question is whether the score's top-ranked cases include plausible rhetorical-function examples and what kinds of false positives it also finds.

The audit pattern matters: top-ranked candidates from the broad target family usually survived the audit, while repetition controls didn't. Exact repetition gives strong surface balance, so it functions as the control case.

The audit also clarified the boundary of the category. Bare identity, apposition, dialogue repair, and exact repetition can be rhetorically relevant apart from isocolon. Conversely, weak length balance can still matter when the two units share a frame strongly enough to invite slot substitution. The coding decisions treat isocolon as a component of a broader rhetorical-function judgment, with false-positive classes instead of a mechanically recoverable surface pattern. Appendix C reports the full stratum counts. Four examples show the main audit distinctions.

The slot-substitution case in (3) is a clear true positive: the frame is repeated and the contrasting material fills the same slot.

(3) a. *You can use this on twist outs.*
    b. *You can use this on braid outs.*

The adversative-contrast case in (4) gives the cleaner non-enumerative pattern: opposed orientations appear in matched phrasing.

(4) a. *so that the broad sides of the cuttings face east and west*
    b. *while the slim sides face north and south.*

The repetition case in (5) is the false-positive control. It's maximally balanced on the surface and was coded as exact repetition, not as isocolon.

(5) a. *Get your feet off of there.*
    b. *Get your feet off of there.*

The label-mismatch case in (6) has an *adversative-antithesis* relation label, but it lacks the balanced opposed cola of classical antithesis.

(6) a. *We are not setting out to turn them into programmers or data crunchers,*
    b. *but want to boost their knowledge level*

7 Interpretation

The findings support a narrow version of the rhetorical-function claim. Balanced form can help present two units as comparable, but it isn't an obligatory marker of comparison. Many comparisons won't be balanced, and many balanced pairs won't be rhetorically interesting. That view predicts an uneven distribution: balance should cluster where matched presentation has rhetorical value and remain absent in many ordinary cases. The corpus result fits that prediction unevenly. Lists and disjunctions supply the largest effects for the broad target family; adversative contrast supplies the cleaner non-enumerative case.

Comparable set names a broad descriptive pattern, not a single classical figure. Two discourse units can be presented as members of the same set as alternatives, contrastive counterparts, or items in a list. The weakest case is only co-enumeration: the units are listed together, and the list makes them co-members without forcing a stronger comparison.

That distinction matters for the interpretation. The list and disjunction effects show that formal balance clusters in co-member structures, but they don't by themselves show that every list is a comparison for its own sake. The cleaner non-enumerative result is the *adversative-contrast* effect, especially in the adversative-only comparison.

The subscore results narrow the claim further. The strongest controlled signal is parison: similarity in the sequence of lexical categories, dependency relations, and related syntactic profiles. Isocolon, understood narrowly as length balance, appears most clearly in the mean-only length model. The effect is weaker for relation labels that look rhetorically relevant in the taxonomy but don't necessarily require balanced presentation in the text.

In particular, *adversative-antithesis* as an eRST relation label only partly overlaps with antithesis in the classical theory of figures. The label *adversative-antithesis* keeps the name while losing the balance. The conservative conclusion leaves antithesis itself intact: the corpus label *adversative-antithesis* serves as a poor direct proxy for the classical figure.

That point remains a limitation as well as a finding. The 80-pair audit contains only eight rows with the *adversative-antithesis* label. These rows support the label-mismatch interpretation, but they're too few to separate label mismatch from a second possibility: genuine semantic antitheses may be expressed through opposition instead of lexical overlap or surface parallelism.

A targeted follow-up will code all 343 eRST *adversative-antithesis* rows for three judgments: classical antithesis, semantic opposition, and parallel opposition. Until then, the near-zero *adversative-antithesis* result is evidence about the corpus label, not a settled test of classical antithesis itself.

The composite formal-balance score tracks matched presentation across several surface dimensions. That matched presentation can occur in contrast, co-ordination, lists, or alternatives. In the list-like cases, grammar and layout already encourage similar units, so the rhetorical interpretation is weaker. Some adversative relations go the other way: they have little balance because their function is concession, qualification, or discourse management instead of matched presentation.

Corpus data record ordinary production, not an ideal of trained rhetorical practice. The evidence shows where balance appears in this corpus. It doesn't show where balanced form could improve a text. The language scope is also narrow. These are English corpus data, mostly from GUM, with English word order, English morphology, English punctuation, and English written and spoken segmentation. Nothing here licenses a claim about how isocolon or parison behave in Ancient Greek, Modern Japanese, or any other language whose prosody, morphology, discourse-unit segmentation, or rhetorical training traditions may differ.

The co-ordination caveat matters. Lists and disjunctions place their units in similar syntactic slots, so parallel syntax is grammatically cheap there. The effect for the broad target family can't be read as a pure rhetorical choice. The joint-only robustness check shrinks the effect, which is what that objection predicts. The effect remains positive, and *adversative-contrast* remains positive inside the adversative family and under a stratified shuffled-label baseline.

Causal interpretation needs the same caution. Relation labels don't cause formal balance. They mark discourse contexts where several sources of balance can converge. In adversative contrast, a writer may choose matched form to present two units as counterparts. In lists and disjunctions, syntax and layout may make matched form easy before a separate rhetorical choice enters. In templatic cases, a local construction may supply both the relation and the balanced wording. The corpus result is compatible with all three stories.

The safer interpretation says that comparable-set relations attract formal balance beyond genre, document, length, and punctuation differences, while much of that balance follows the syntax of co-ordination. Relation labels don't guarantee classical figures, and classical figures needn't appear where relation labels promise them.

The pattern also clarifies the scope of the corpus study. The evidence stays textual and distributional. It concerns where balanced form occurs, not how readers experience those pairs. A high composite formal-balance score by itself still falls short of a rhetorical-figure label. The analysis shows that the composite formal-balance score is systematically higher in particular discourse-relation contexts. The human audit illustrates why many top-ranked cases are rhetorically relevant, but it remains pilot evidence. A stronger validation would require an independent annotation layer: a new set of human-coded rhetorical-function labels.

The RST Signalling Corpus isn't part of the analysis yet because access is still pending. Once available, that corpus can add a later triangulation point.

Work on discourse signalling has shown that coherence relations are often signalled by more than discourse markers, including lexical, syntactic, graphical, semantic, and genre-based cues. That point is developed in the original signalling annotation work and in later studies of RST signals and multiple signals (Das & Taboada, 2018a, 2018b, 2019; Taboada & Das, 2013). This literature gives the project a stronger comparative baseline. If formal balance behaves like a discourse signal, then it should be testable alongside other signal types in the discourse-signalling inventory.

8 Conclusion

A corpus study can't prove that isocolon makes readers perceive comparison as more forceful. It can test a cleaner textual claim: writers use formal balance unevenly across discourse relations, and the relations most associated with comparable-set construals should show more of it. These results support that claim for the broad target family of comparison-like and co-ordinative relations, with the largest effects in lists and disjunctions and the cleaner non-enumerative effect in adversative contrast.

The analysis gives selective support to ANTITHESIS as an inherited figure label. The effect for the narrow adversative family stays small, and the discourse-relation label *adversative-antithesis* sits close to zero. That finding sharpens the mapping between relation labels and classical figures. The strongest signal comes from parison-like parallelism. Isocolon narrowly understood as length balance contributes less to the main effect for the composite formal-balance score. Balanced form remains a rhetorical resource, and the corpus evidence supports a claim about comparable-set construals and formal balance more securely than a claim about equal cola alone.

The robustness checks sharpen that conclusion. Length balance appears under mean-length conditioning without maximum length, while parison leads the main model for the composite formal-balance score. The joint-only comparison also shows that a large share of the effect for the broad target family comes from co-ordination itself. The stratified permutation check places the observed effects above document, length, punctuation, and explicitness-matched shuffled-label baselines. The result maps where measurable figural form becomes distributionally useful.

Data and code availability

The analysis scripts, coding materials, and derived audit summaries are kept in the project repository: https://github.com/BrettRey/isocolon-and-comparative-construal. The raw GUM/eRST source files are excluded from the repository; the scripts expect a local corpus checkout under `data/raw/gum-erst`. The RST Signalling Corpus isn't included yet because access is still pending.

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A Extraction diagnostics

The derived DISRPT relation table has relation rows and token intervals, not the full eRST graph-node inventory. The counts in Table 4 are interval-level checks. Gapped rows have material between the two linked units. Interleaved rows have token intervals that overlap or cross in a way that prevents a clean left-unit/right-unit comparison. Discontinuous adjacent rows remain in the main analysis; the robustness checks include a continuous-span-only subset.

Extraction diagnostic	Count
Valid explicit/implicit relation rows	34,976
Adjacent relation rows used in analysis	17,456
Non-adjacent relation rows excluded	17,520
gapped rows excluded	15,349
interleaved rows excluded	2,171
Adjacent rows with discontinuous token intervals	1,963
Exact duplicate adjacent span pairs	0
Adjacent explicit rows	4,943
Adjacent implicit rows	12,513

Table 4: Extraction diagnostics for the adjacent-pair analysis. Counts come from token intervals in the DISRPT relation export. Exact duplicate span pairs are counted by document, unit-1 tokens, and unit-2 tokens.

B Validation details

Figure 3 shows the precision/recall tradeoff for score-thresholded inspection slices. The two panels use the same numerator with different denominators. In Panel A, the 97.5th-percentile point divides 24 *syn-prl* rows by 438 inspected rows. In Panel B, the same point divides those 24 rows by all 73 *syn-prl* rows in the corpus.

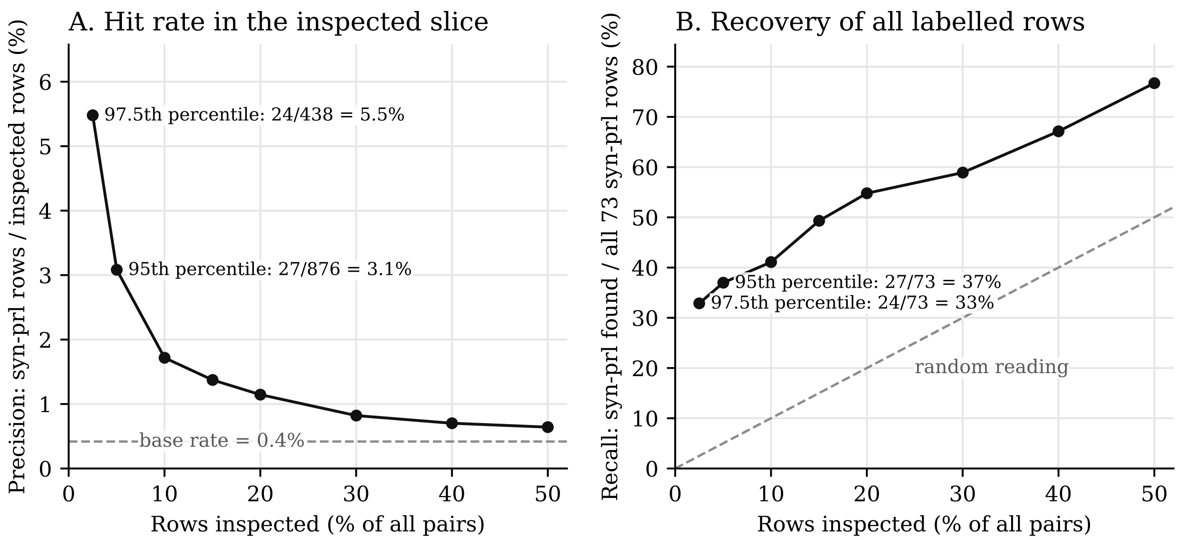

Figure 3: Validation enrichment for top-scored slices. Panel A shows precision: *syn-prl* rows divided by inspected rows. The dashed line is the corpus base rate. Panel B shows recall: recovered *syn-prl* rows divided by all 73 *syn-prl* rows in the corpus. The dashed diagonal shows the recovery expected from reading a random slice of the same size. Labels mark the 95th- and 97.5th-percentile score cutoffs; tied scores at a cutoff are included. The figure checks ranking utility, not rhetorical function.

Measure	AUC	Labelled mean	Unlabelled mean
Composite formal balance	0.73 [0.66, 0.80]	0.50	0.33
Isocolon	0.65 [0.58, 0.72]	0.68	0.56
Parison	0.75 [0.69, 0.82]	0.44	0.21
Lexical echo	0.72 [0.65, 0.79]	0.20	0.06

Table 5: Validation against the 73 eRST *syn-prl* signal labels in 17,456 adjacent relation pairs. AUC gives the chance that a random labelled pair receives a higher value than a random unlabelled pair. The labelled-mean and unlabelled-mean columns are the average values for pairs that do and don't carry the label, on the same 0–1 scale as the composite formal-balance score and the three subscores. Brackets give approximate Hanley–McNeil 95% intervals for AUC.

C Qualitative audit strata

Sample stratum	n	Yes	No	Uncertain	Main issue
High broad target family	16	14	2	0	Slot substitution
Mid broad target family	12	7	4	1	Weak frames
Narrow adversative family	14	6	8	0	Label mismatch
High non-target/artifact	19	4	12	3	Dialogue/apposition
Concession candidates	5	1	4	0	Asymmetry
Repetition controls	9	0	9	0	Exact identity
Baseline non-target	5	1	4	0	Weak or no frame

Table 6: Human qualitative audit summary for 80 candidate pairs.

D Robustness details

Each check removes one easy explanation for the main result and asks whether the effect survives. If balance only looked elevated because of some background difference between target and non-target pairs, then matching or stripping out that difference should make the effect shrink toward zero. Table 7 gives selected checks.

Check	Target and outcome	Estimate (points)	Clustered 95% interval
Main	Broad target family, composite	4.1	[3.6, 4.7]
Main	*joint-list* composite	4.4	[3.7, 5.1]
Main	*adversative-contrast* composite	2.4	[1.3, 3.5]
Mean-length only	Broad isocolon	6.6	[5.6, 7.7]
Mean-length only	*adversative-contrast* isocolon	7.3	[5.0, 9.6]
Joint-only	List/disjunction composite	1.7	[0.6, 2.8]
Joint-only	List/disjunction parison	2.9	[1.0, 4.8]
Adversative-only	Contrast composite	3.0	[1.4, 4.7]
Adversative-only	Contrast parison	5.9	[3.0, 8.8]
GUM-only	Broad target family, composite	4.0	[3.4, 4.6]
Continuous spans only	Broad target family, composite	4.5	[3.8, 5.1]

Table 7: Selected robustness checks, reported as score-point differences on a 0–100 scale. The joint-only rows compare *joint-list* and *joint-disjunction* against other *joint-** relations. Mean-length-only rows control log mean length and omit log maximum length. The adversative-only rows compare *adversative-contrast* against other adversative relations. The continuous-spans-only row excludes adjacent rows where either discourse unit has a comma-separated token interval.

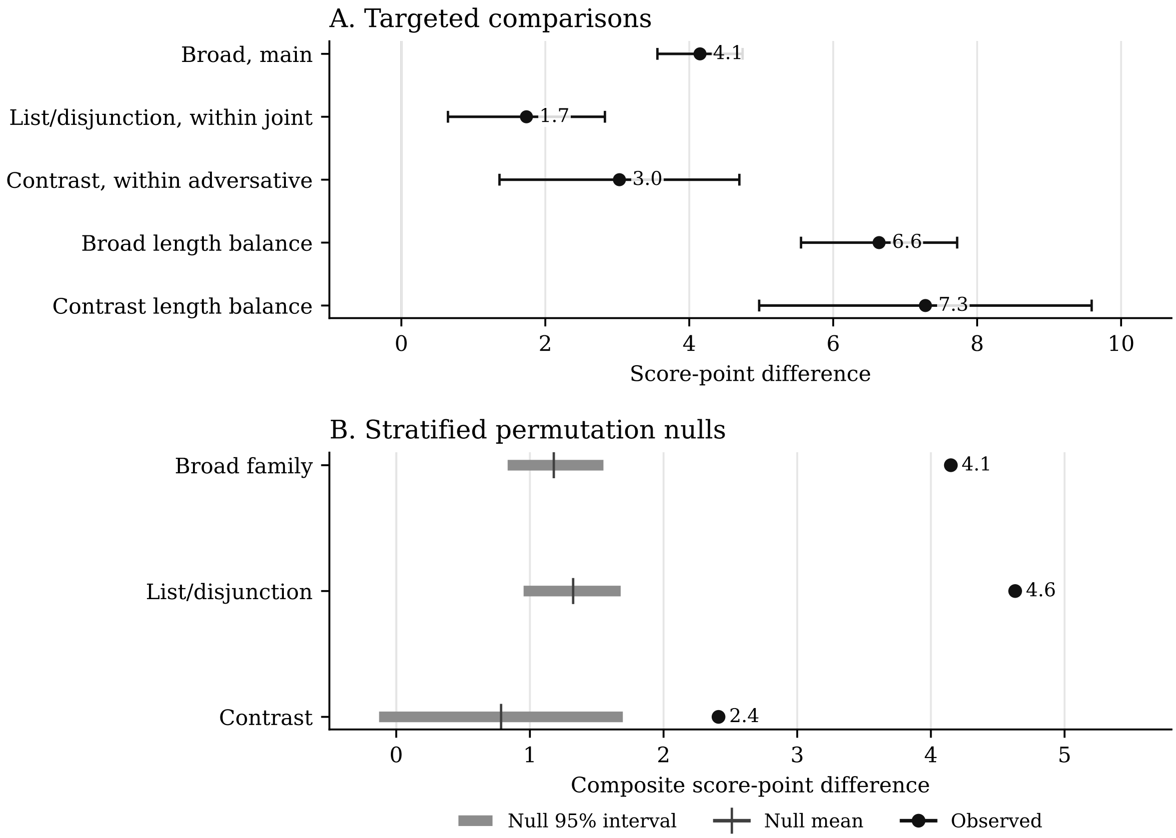

Figure 4: Targeted robustness checks and stratified permutation nulls, reported as score-point differences on a 0–100 scale. Panel A shows adjusted differences with document-clustered 95% intervals. Panel B compares observed composite effects with the null mean and central 95% interval from stratified target-label shuffles; here the null means the effect expected if target labels carried no relation-specific information inside the matched strata.

The composite weights were also varied across 66 nonnegative alternative weightings of the three subscores. The effect for the broad target family stays positive, with an uncertainty interval that excludes zero, in every one of the 66. For *adversative-contrast*, 65 of the 66 are positive and 59 have uncertainty intervals that exclude zero.

E Stratified permutation nulls

The permutation check asks whether the effect could be an accident of which relations happen to wear the target label. It holds the texts, lengths, punctuation, and explicitness fixed, then shuffles target labels inside matched strata. The strata are defined by document, explicitness, same-sentence status, colon and semicolon punctuation, and quartile of mean word length.

Target	Observed	Null mean	Null 95%	Strata	Rows	Target rows	p
Broad target family	4.1	1.2	[0.8, 1.6]	1,240	8,277	2,174	0.002
List/disjunction	4.6	1.3	[1.0, 1.7]	1,033	7,047	1,659	0.002
adversative-contrast	2.4	0.8	[-0.1, 1.7]	201	1,472	220	0.002

Table 8: Stratified permutation nulls for the composite formal-balance effect, reported as score-point differences on a 0–100 scale. Target labels are shuffled within document, explicitness, same-sentence status, colon/semicolon status, and mean-length quartile strata. The null mean is the average effect after that relabelling. The strata, rows, and target-rows columns report the portions of the data that could move under the stratified shuffle. The p values are two-sided empirical values from 500 permutations, with the standard plus-one correction, and should be read at that resolution.

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